

Initial Experiments: Extending the MVP Simulator

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Extends the MVP simulator (`docs/mvp/mvp_specification.tex`) by enriching the state space, action space, and reward function, moving toward the full MDP defined in `docs/design/initial_design.tex`.

1 State Space Extension

References `docs/design/initial_design.tex` §3 and Issue #124.

1.1 Enriched StateView

1.2 Contextual Bandit Conditioning

1.3 Open Questions

2 Action Space Extension

References `docs/design/initial_design.tex` §3 and Issue #126.

2.1 ActionConfig

2.2 Transition Model Updates

2.3 Agent Compatibility

2.4 Open Questions

3 Reward Function Extension

References `docs/design/initial_design.tex` §3 and Issue #132. *Implementation: PR 1c (feat/1c-reward) — merged.*

3.1 RewardWeightsConfig

An optional `reward_weights` block controls the delayed reward component. When absent the reward is purely per-state (MVP backward-compatible mode):

$$r(s, a, t) = m[t \bmod S] \cdot r(s')$$

When present, the schema accepts these fields:

Field	Type	Default
mode	Literal["multi_timescale"]	"multi_timescale"
delayed_reward_interval	int ≥ 1	21
delayed_reward_value	float ≥ 0	10.0
delayed_reward_scale	float None	None
delayed_reward_threshold	float $\in [0,1]$ None	None

All numeric fields are validated at load time via Pydantic Field constraints. The mode is constrained to Literal["multi_timescale"]; when additional modes are added later the union can be expanded.

3.2 Flat Bonus (Option A)

When `delayed_reward_scale` is None a flat bonus fires at every interval boundary:

$$r_{\text{bonus}} = \begin{cases} \text{value} & \text{if } t > 0 \text{ and } t \equiv 0 \pmod{N} \\ 0 & \text{otherwise} \end{cases}$$

where $N = \text{delayed_reward_interval}$ and t is the zero-indexed cumulative step count (`state.global_step`).

3.3 Scaled Bonus (Option B)

When `delayed_reward_scale` is set the bonus depends on the fraction of steps in the interval where the simulated user was in the active state:

$$\text{rate} = \frac{\text{active_count}}{N} \quad r_{\text{bonus}} = \begin{cases} \text{scale} \times \text{rate} & \text{if } \text{rate} \geq \text{threshold} \\ 0 & \text{otherwise} \end{cases}$$

The active count is tracked in `CompoundReward._active_count`, incremented on every `reward()` call where `state.activity == "active"` and reset to zero at each interval boundary.

3.4 Episode Boundaries

The `RewardHandler ABC` now defines a `reset()` method that clears episode-local state (`_active_count`). `Environment.reset()` calls `self._reward.reset()` so that reusing a reward instance across episodes does not leak state.

3.5 Results

The 1c experiment config is at `docs/initial_experiments/configs/1c_reward_only.yaml`. Total rewards for Thompson Sampling over one run with seed 42:

Mode	Total reward	Bonus firings
No bonus (MVP baseline)	165.0	0
Flat bonus (interval=21, value=10.0)	377.0	22
Scaled bonus (scale=5.0, threshold=0.6)	168.1	0

The flat bonus fires 21–22 times over the 450-step episode ($450/21 \approx 21.4$), contributing roughly $22 \times 10.0 = 220$ in additional reward. The scaled bonus with `threshold = 0.6` never fires for this configuration: the Thompson Sampling agent cannot sustain a 60% active rate within a 21-step interval given the hand-tuned transition probabilities. Lowering the threshold or increasing the action efficacy would be needed for the scaled bonus to take effect.

3.6 Open Questions

- The `delayed_reward_interval` and `delayed_reward_value` parameters are placeholders. Calibration against real-world user data would inform clinically meaningful values.
- The scaled bonus threshold (default 0.6) and scale (default 5.0) were chosen arbitrarily. A grid search could identify regions where the delayed reward shapes agent behaviour without overwhelming the per-step signal.
- The flat vs. scaled choice is a config toggle today. A future extension could blend both: a fixed bonus plus a scaled bonus for users who meet the threshold.

4 Combined Results

Results from the pairwise and full-integration experiments will appear here once all three extension PRs (1a, 1b, 1c) have been merged and the integration PRs (4–7) are complete. The config matrix of seven experiment configurations is defined in `docs/initial_experiments/plans/state_action_reward_epic.md`. Currently available:

- `docs/initial_experiments/configs/1c_reward_only.yaml` (PR 1c, 2 actions, 2 states, multi-timescale reward) — see §3.